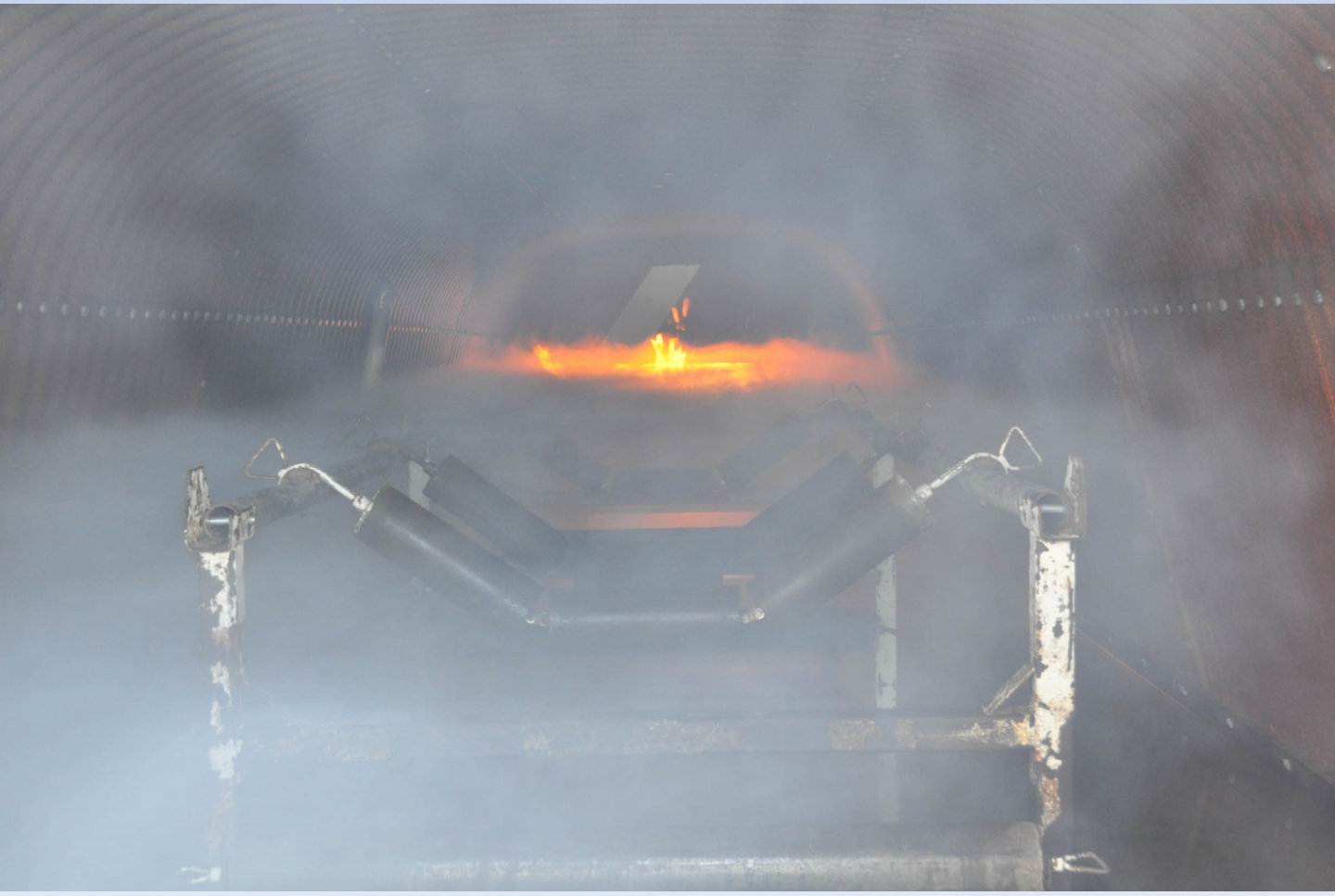
2015 MTTC Pennsylvania Regional Mine Rescue Skills Contest



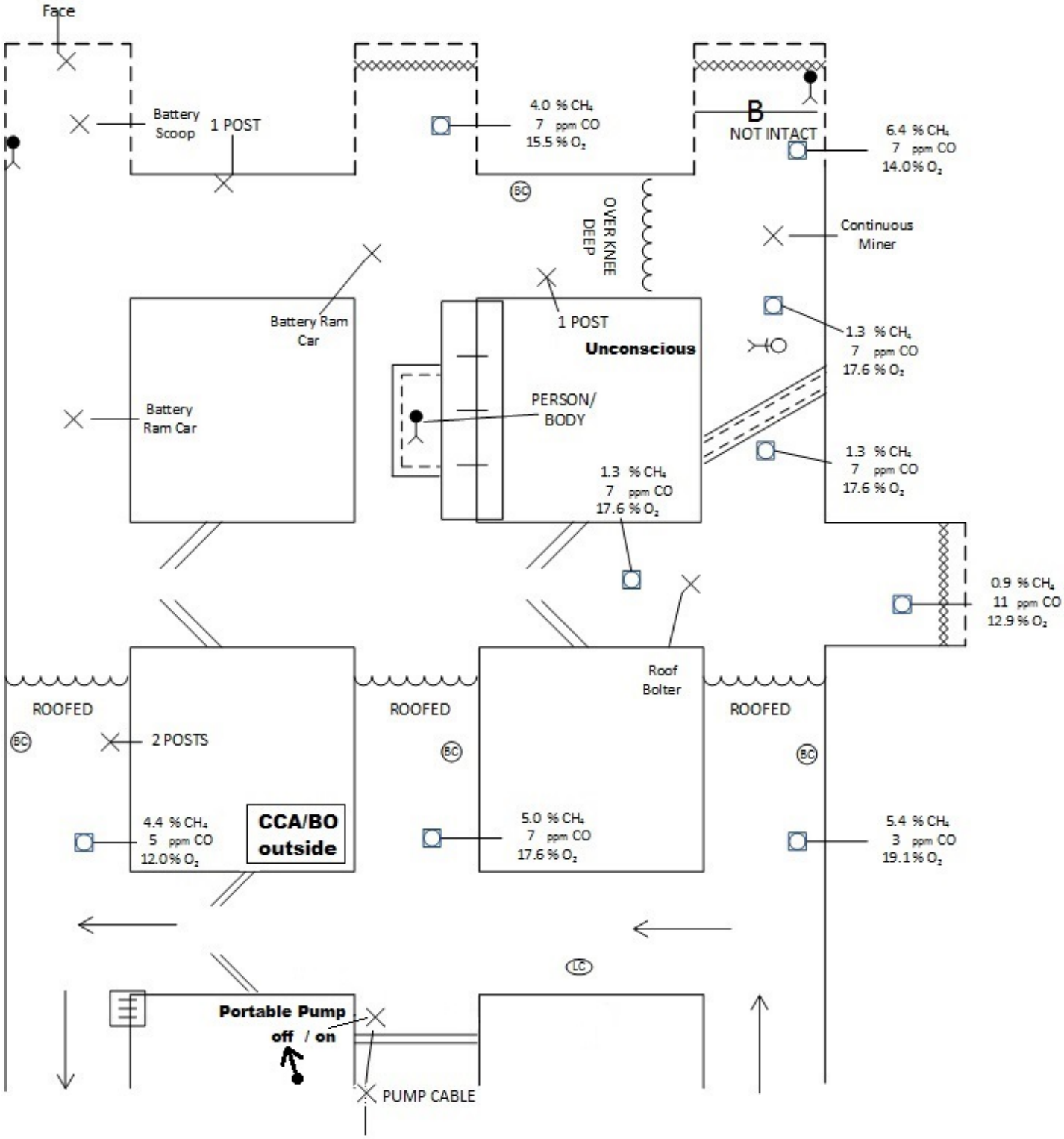
April 21-25
Ruff Creek, PA

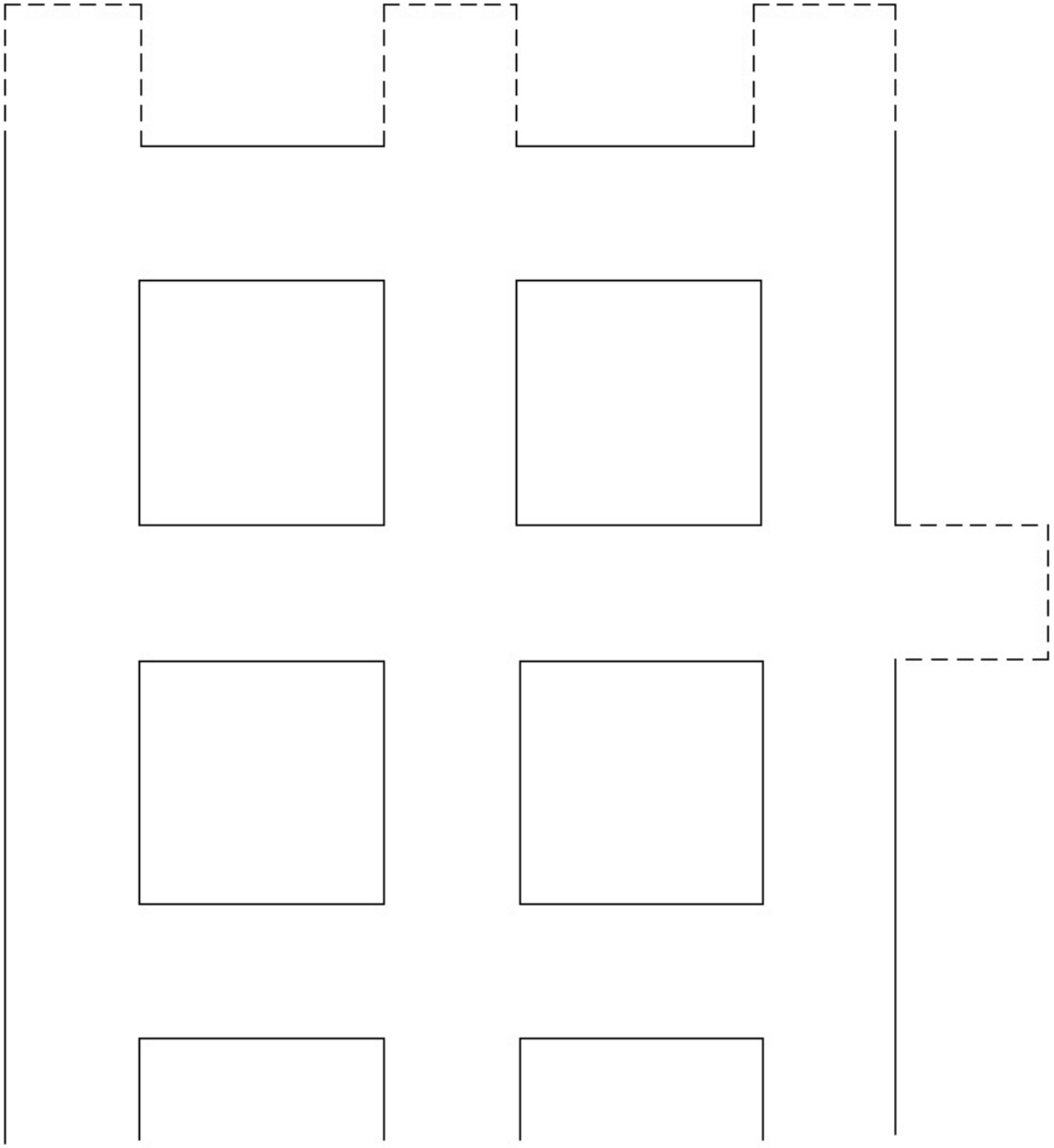
Problem Statement

You are located in the first crosscut underground of the M&M # 1 Mine. A fresh air base was established here when a responsible person went underground and found water roofed in all three entries.

We believe the 4 right section may have cut into old works. We have been calling the section but none of the four miners have answered.

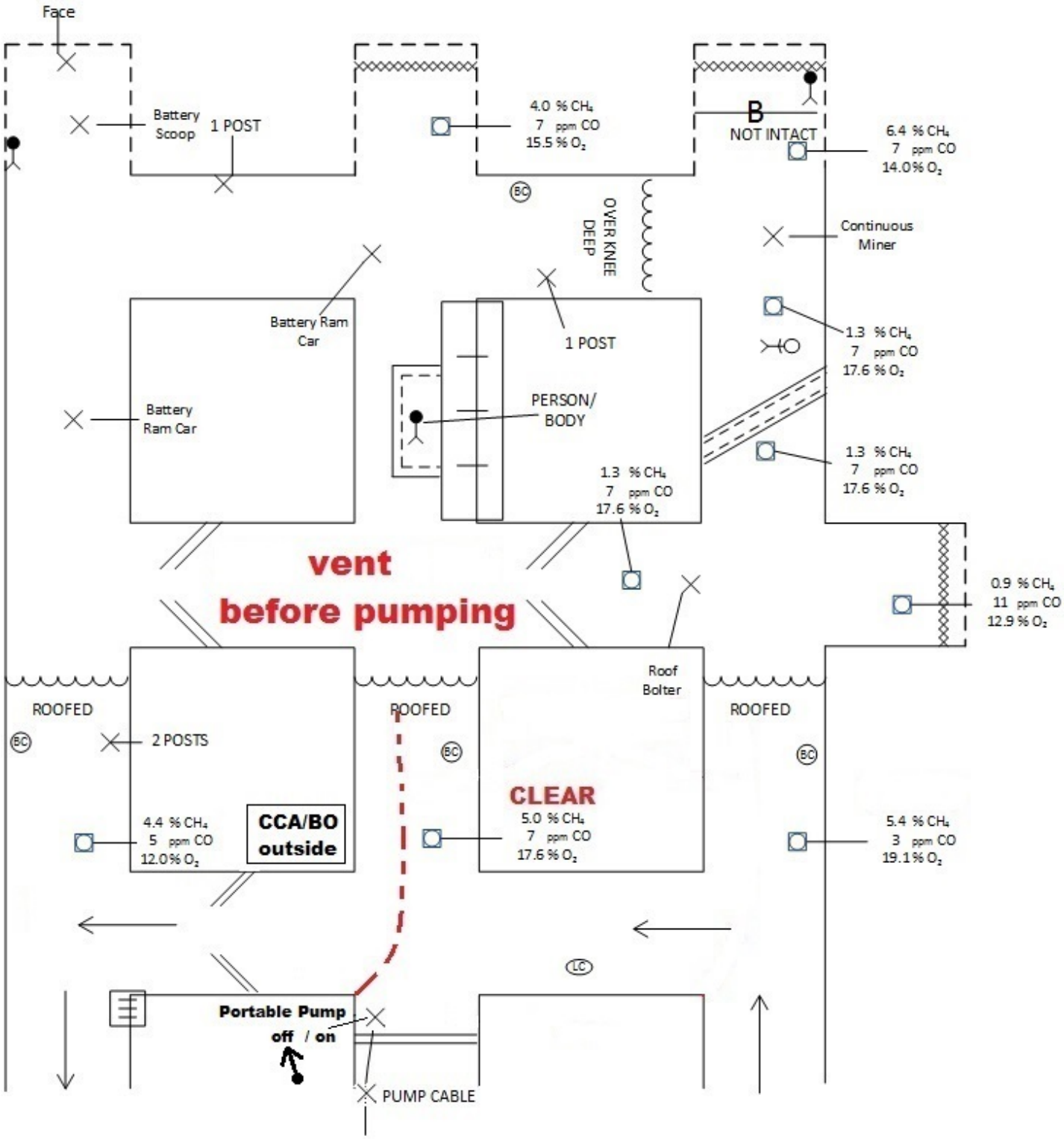
There is a trained mine rescue team here to back you up and power inby the fresh air base has been locked out. The maps you will receive when you start the clock are not up to date.

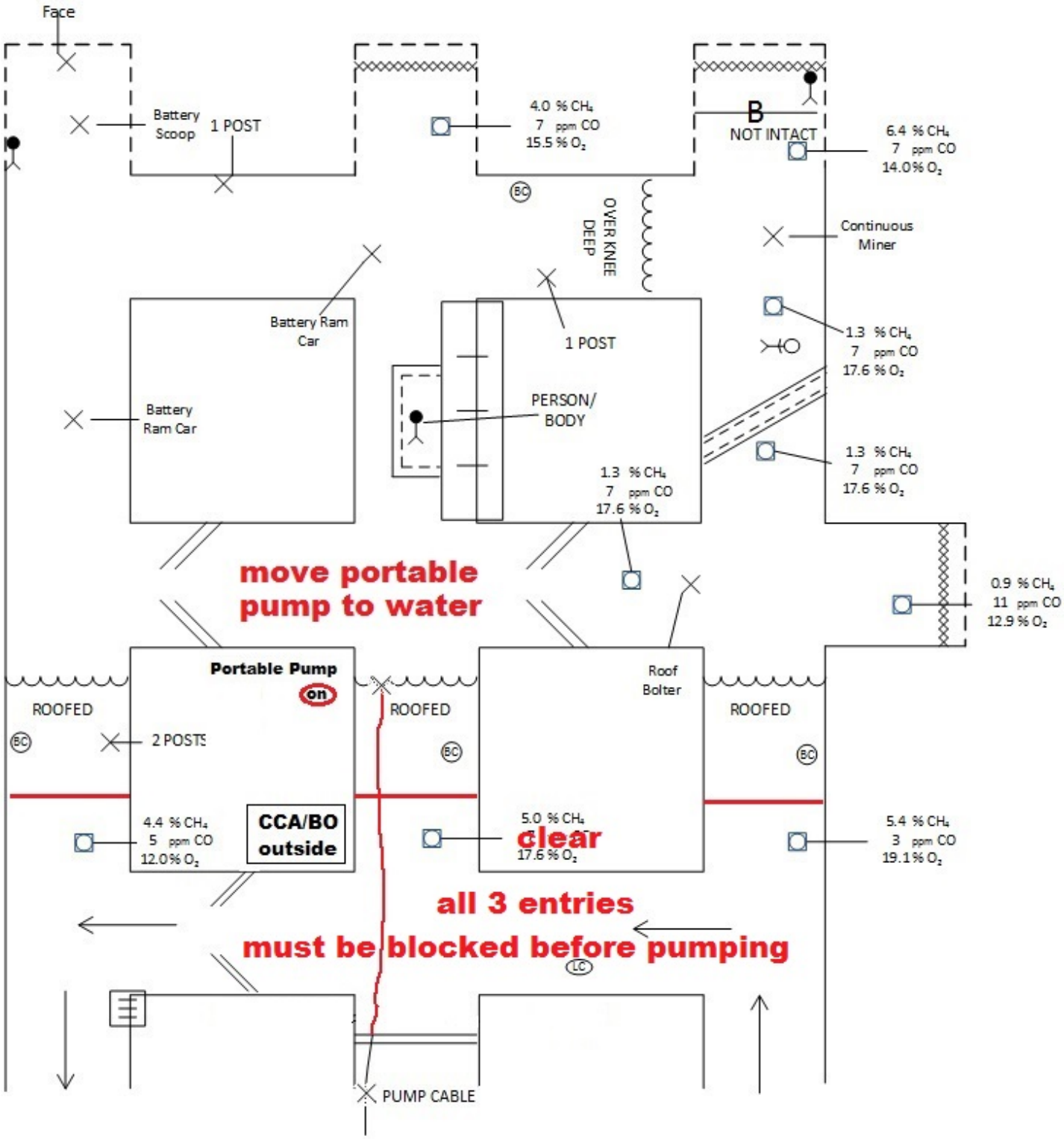

Face
Battery Scoop
1 POST
4.0 % CH4
7 ppm CO
15.5 % O2
B
NOT INTACT
6.4 % CH4
7 ppm CO
14.0% O2
BC
OVER KNEE DEEP
Continuous Miner
Battery Ram Car
1 POST
Unconscious
PERSON/ BODY
1.3 % CH4
7 ppm CO
17.6 % O2
Battery Ram Car
1.3 % CH4
7 ppm CO
17.6 % O2
1.3 % CH4
7 ppm CO
17.6 % O2
0.9 % CH4
11 ppm CO
12.9 % O2
ROOFED
ROOFED
ROOFED
Roof Bolter
BC
2 POSTS
BC
BC
4.4 % CH4
5 ppm CO
12.0% O2
CCA/BO outside
5.0 % CH4
7 ppm CO
17.6 % O2
5.4 % CH4
3 ppm CO
19.1% O2
LC
Portable Pump
off / on
PUMP CABLE

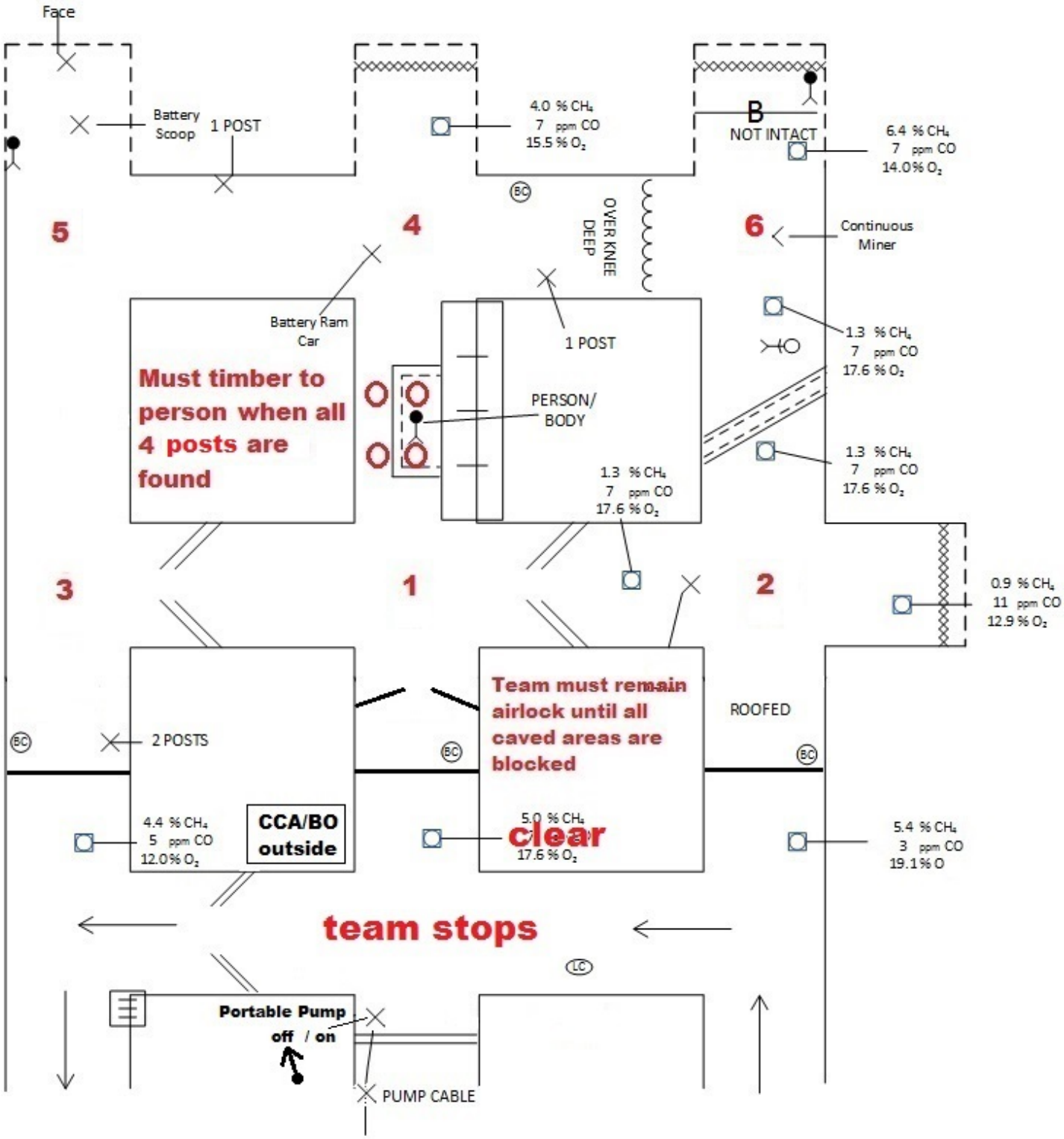


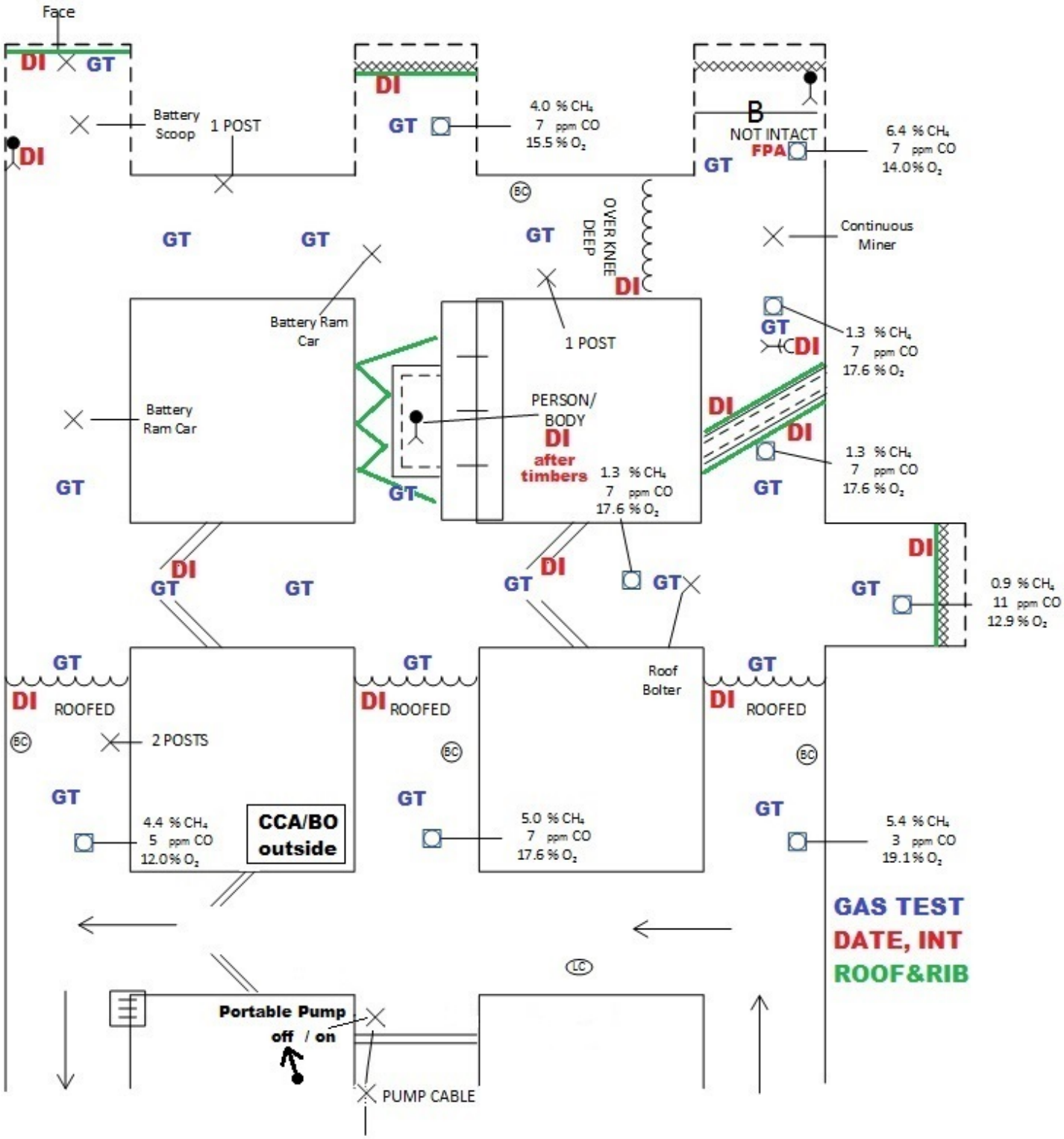
Written Instructions

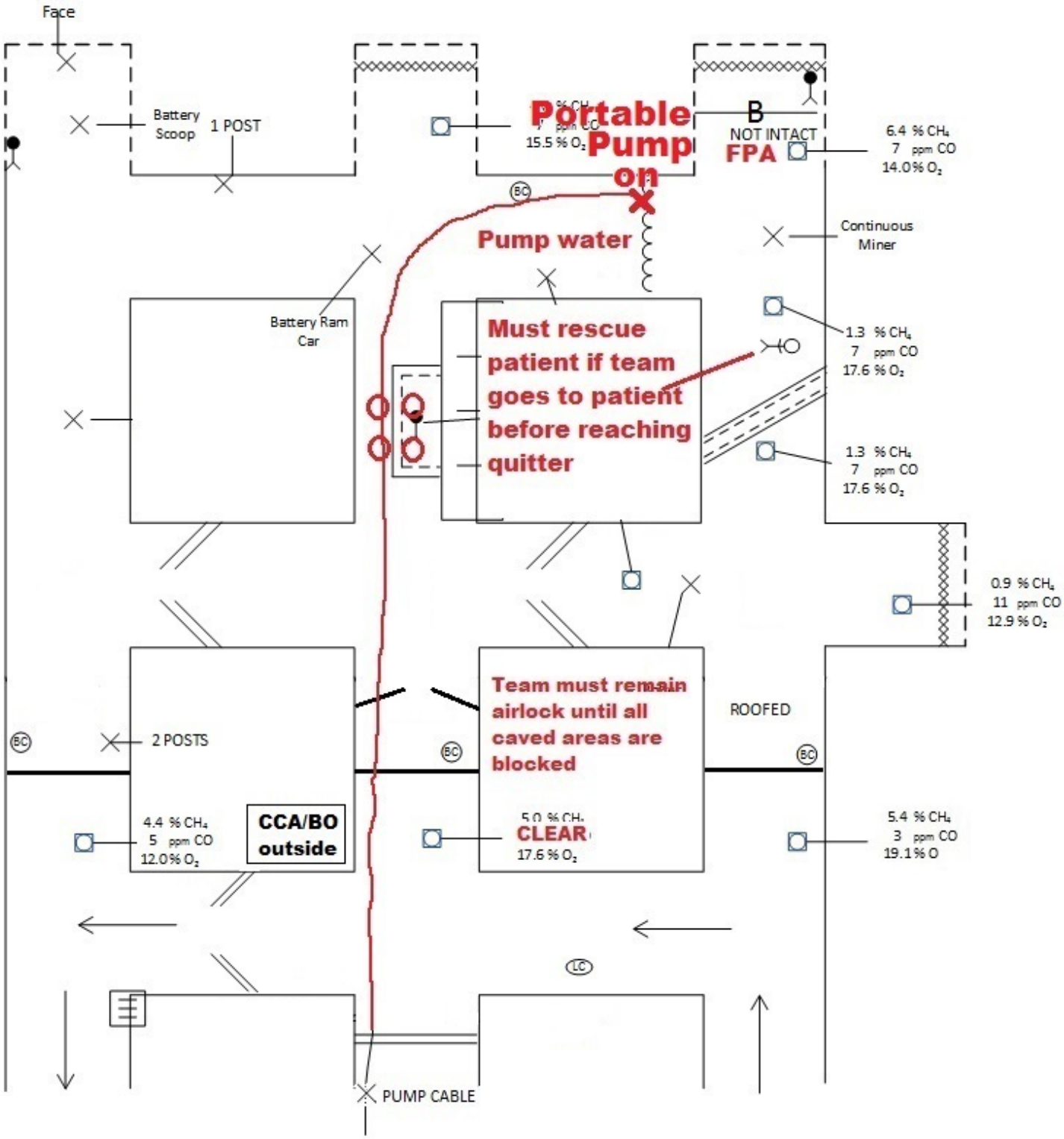
- Account for all missing miners and bring live persons to the fresh air base
- There is a blowing fan that is on and can't be shut off
- Portable pump can be turned on or off by the team
- Water pumped in the no. 2 entry just inby fresh air base (FAB) will also remove water in no.1 and no. 3 entry and connecting crosscut
- Team must use the no.2 entry on the intital exploration inby the FAB
- The Command Center Attendant / BO is outside


Face
Battery Scoop
1 POST
4.0 % CH4
7 ppm CO
15.5 % O2
B
NOT INTACT
6.4 % CH4
7 ppm CO
14.0% O2
BC
OVER KNEE DEEP
Continuous Miner
Battery Ram Car
1 POST
1.3 % CH4
7 ppm CO
17.6 % O2
Battery Ram Car
PERSON/ BODY
1.3 % CH4
7 ppm CO
17.6 % O2
vent
before pumping
0.9 % CH4
11 ppm CO
12.9 % O2
ROOFED
ROOFED
Roof Bolter
ROOFED
BC
2 POSTS
BC
BC
CLEAR
5.0 % CH4
7 ppm CO
17.6 % O2
4.4 % CH4
5 ppm CO
12.0% O2
CCA/BO outside
5.4 % CH4
3 ppm CO
19.1% O2
LC
Portable Pump
off / on
PUMP CABLE


Face
Battery Scoop
1 POST
4.0 % CH4
7 ppm CO
15.5 % O2
B
NOT INTACT
6.4 % CH4
7 ppm CO
14.0% O2
BC
OVER KNEE DEEP
Continuous Miner
Battery Ram Car
1 POST
PERSON/ BODY
1.3 % CH4
7 ppm CO
17.6 % O2
Battery Ram Car
1.3 % CH4
7 ppm CO
17.6 % O2
1.3 % CH4
7 ppm CO
17.6 % O2
move portable pump to water
0.9 % CH4
11 ppm CO
12.9 % O2
Portable Pump
on
Roof Bolter
ROOFED
ROOFED
ROOFED
BC
2 POSTS
BC
BC
4.4 % CH4
5 ppm CO
12.0% O2
CCA/BO outside
5.0 % CH4
clear
17.6 % O2
5.4 % CH4
3 ppm CO
19.1% O2
all 3 entries
must be blocked before pumping
LC
PUMP CABLE


Face
Battery Scoop
1 POST
4.0 % CH4
7 ppm CO
15.5 % O2
B
NOT INTACT
6.4 % CH4
7 ppm CO
14.0% O2
BC
OVER KNEE DEEP
5
4
6
Continuous Miner
Battery Ram Car
Must timber to person when all 4 posts are found
1 POST
PERSON/ BODY
1.3 % CH4
7 ppm CO
17.6 % O2
1.3 % CH4
7 ppm CO
17.6 % O2
1.3 % CH4
7 ppm CO
17.6 % O2
3
1
2
0.9 % CH4
11 ppm CO
12.9 % O2
Team must remain airlock until all caved areas are blocked
ROOFED
BC
2 POSTS
BC
BC
4.4 % CH4
5 ppm CO
12.0% O2
CCA/BO outside
5.0 % CH4
clear
17.6 % O2
5.4 % CH4
3 ppm CO
19.1% O
team stops
LC
Portable Pump
off / on
PUMP CABLE


Face
DI
GT
Battery Scoop
1 POST
4.0 % CH4
7 ppm CO
15.5 % O2
B
NOT INTACT
FPA
6.4 % CH4
7 ppm CO
14.0% O2
BC
OVER KNEE DEEP
Continuous Miner
Battery Ram Car
1 POST
PERSON/ BODY
after timbers
1.3 % CH4
7 ppm CO
17.6 % O2
0.9 % CH4
11 ppm CO
12.9 % O2
ROOFED
2 POSTS
Roof Bolter
4.4 % CH4
5 ppm CO
12.0% O2
CCA/BO outside
5.0 % CH4
7 ppm CO
17.6 % O2
5.4 % CH4
3 ppm CO
19.1% O2
GAS TEST
DATE, INT
ROOF&RIB
LC
Portable Pump
off / on
PUMP CABLE


Face
Battery Scoop
1 POST
% CH
ppm CO
15.5 % O2
Portable Pump on
B
NOT INTACT
FPA
6.4 % CH4
7 ppm CO
14.0% O2
BC
Pump water
Continuous Miner
Battery Ram Car
Must rescue patient if team goes to patient before reaching quitter
1.3 % CH4
7 ppm CO
17.6 % O2
1.3 % CH4
7 ppm CO
17.6 % O2
0.9 % CH4
11 ppm CO
12.9 % O2
Team must remain airlock until all caved areas are blocked
ROOFED
2 POSTS
4.4 % CH4
5 ppm CO
12.0% O2
CCA/BO outside
5.0 % CH4
CLEAR
17.6 % O2
5.4 % CH4
3 ppm CO
19.1% O
LC
PUMP CABLE

Keys to Problem

- Upon completion of the team checking all openings to the area to be explored Team must wing the explosive mixture in the no. 2 entry before pumping water roofed (See vent)
- Team will then airlock across no.1 and no. 3 entry and take two brattice cloths and portable pump up to the water roofed in the no. 2 entry
- Team will build airlock and then pump water roofed in the no. 2 entry
- Team must remain airlocked until all openings inby the FAB are blocked or airlock out each time team goes to/from FAB
- Team must advance toward no.3 entry after finding contaminant in crosscut between 2 and 3 entry at team stop 1
- Team must advance up the no.2 entry into last open crosscut because of contaminant found just outby diagonal unsafe roof in the no. 3 entry
- Team must timber to person after finding 4 posts at team stop 4
- After team stop 5, team can move portable pump up the no. 2 entry into last open crosscut and pump water over knee deep to find missing miners
- Team must FPA when they find the explosive mixture in the no. 3 entry after pumping water over knee deep between 2 and 3 entry. CO 11 PPM is going into caved area in first crosscut in no. 3 entry and map not up to date
- Team must not go to Barricade inby explosive mixture in the no. 3 entry
- If team goes to unconscious patient before going to explosive mixture inby intersection in the no. 3 entry, Team can rescue unconscious patient and FPA at explosive mixture inby intersection and then return to the FAB
- If team goes to explosive mixture inby intersection in no.3 entry before going to patient outby intersection, team must FPA where they have been and not go to unconscious patient and then return to the FAB